

10.3.4. Discrete - Bitmap

Bitmaps SHALL be encoded using the TLV unsigned integer type, with the width selected automatically at runtime based on the actual value.

10.3.5. Composite - String

While strings are a derived data type, they SHALL be encoded using the [TLV UTF-8 string](#) type.

10.3.6. Composite - Octet String

Octet strings SHALL be encoded using TLV Byte Strings.

10.3.7. Collection - Struct

Structure types in schema SHALL be encoded using the TLV structure type.

10.3.8. Collection - List

The entirety of a list SHALL be encoded as a TLV array.

A list index SHALL start at 0.

Lists SHALL have a maximum size of 65535 elements ($2^{16}-1$).

10.3.9. Derived Types

All derived types (with the exception of strings) SHALL be encoded according to their base type.

10.3.10. Field IDs

Field IDs SHALL be encoded as:

- A context tag when the MEI prefix encodes a standard/scoped source.
- A fully-qualified profile-specific tag when the MEI prefix encodes a manufacturer code. The Vendor ID SHALL be set to the manufacturer code, the profile number set to 0 and the tag number set to the MEI suffix.

NOTE Support for encoding Field IDs with an MC source is provisional.

10.4. Sample Clusters

This section defines sample clusters (with attributes, events, and commands) for illustrative purposes; they SHALL NOT be interpreted as real clusters.

10.4.1. Disco Ball Cluster

This example cluster controls an imaginary mirrored disco ball, for the express purpose of disco